

Extended Kalman Filter Based Speed Sensorless PMSM Control with Load Reconstruction

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Abstract – This paper describes a study and experimental verification of sensorless control of Permanent Magnet Synchronous Motor. This structure bases on the extended Kalman filter theory using only the measurement of the motor current for on-line estimation of speed, rotor position and load torque reconstruction. Control structure such as Kalman filtering, in real time requires a very fast signal processor in special way, adapted to perform complex mathematical calculations. The Digital Signal Processors have become cheaper and their performance greater. Without using position and torque sensors, it has become possible to apply described control structure as a cost-effective solution.

I. INTRODUCTION

Permanent Magnet Synchronous Motor drives are widely used in industrial processes, due to their inherent high torque to inertia ratio, small sizes and enhanced dynamic performances [4]. The Field Oriented Control strategy permits one to fast response to load and speed changes, but it requires information about shaft position [4]. Presented system has led to extreme reduction of mechanical sensors. Extended Kalman Filter [1] has been used for estimation these variables basing on supplied voltages and measurement currents [2,3,5,7]. The system is also expanded to load torque and speed estimation.

II. MODELING OF THE SYSTEM

The mathematical model of object is a major task during building up an estimator. The proper model can simplify a solution of estimation approach. The method described in [4] based on network equations of stator in rotating frame (dq) associated with rotor flux direction:

$$u_d = R_s \cdot i_d + L_d \frac{di_d}{dt} - \omega_r \cdot L_q \cdot i_q, \quad (1)$$

$$u_q = R_s \cdot i_q + L_q \frac{di_q}{dt} + \omega_r \cdot L_d \cdot i_d + \omega_r \cdot \Psi_f, \quad (2)$$

where: u_d, u_q are stator voltages, i_d, i_q – stator currents, L_d, L_q – stator inductances, Ψ_f – rotor flux, ω_r – mechanical speed.

Basing on electromechanical equation:

$$T_{el} = \frac{3}{2} \cdot p \cdot i_q \left(\Psi_f - (L_q - L_d) i_d \right), \quad (3)$$

where: p is a number of pair of poles, and mechanical acceleration of motor shaft:

$$\frac{d\omega_r}{dt} = \frac{1}{J} (T_{el} - T_{load}), \quad (4)$$

it is possible to define general movement equation as

$$\frac{d\omega_r}{dt} = \frac{p}{J} \cdot i_q \left(\Psi_f - (L_q - L_d) i_d \right) - \frac{T_{load}}{J}. \quad (5)$$

Many speed observers described in literature do not recognize load torque [2]. It is assumed, the velocity is treated like constant in short period of time. These solutions avoid load torque which is treated like unknown disturbance. The derivative of angular shaft position is defined by:

$$\frac{d\gamma}{dt} = p\omega_r. \quad (6)$$

Basing on described mathematical formulas, we can write down the state-space vector:

$$x_k = \begin{bmatrix} i_d & i_q & \omega_r & \gamma & T_{load} \end{bmatrix}^T, \quad (7)$$

similar to [5].

System is described by the following state equations:

$$x_{k+1} = \mathbf{A}_k(x_k) \cdot x_k + \mathbf{B}_k(x_k) \cdot u_k + w_k, \quad (8)$$

$$y_k = \mathbf{C}_k(x_k) \cdot x_k + v_k. \quad (9)$$

In this case all variables are available. The system has a electrical input – stator voltages ($\alpha\beta$ reference frame):

$$u_k = \begin{bmatrix} u_\alpha & u_\beta \end{bmatrix}^T, \quad (10)$$

and electrical output – stator currents:

$$y_k = \begin{bmatrix} i_\alpha & i_\beta \end{bmatrix}^T. \quad (11)$$

The value of external torque T_{load} is treated as state variable (7) and part of disturbance vector w_k (8) [1,3,5,7]. This operation allows to estimate parameters of differential equations of model. It is also assumed, that this estimated variable is treated like white noise and it is constant in small interval (sampling time T_s):

$$\frac{dT_{load}}{dt} \approx 0. \quad (12)$$

It is noticed, that the state-space equations are nonlinear. Some elements of the $\mathbf{A}_k$, $\mathbf{B}_k$, $\mathbf{C}_k$ matrices depend on an instant x_k vector (values of angular position γ , shaft velocity ω_r and current i_d). In order to build estimator basing on EKF, the nonlinear model of PMSM is a priority.

There are matrices:

$$\mathbf{A}_k = \begin{bmatrix} 1 - T_s \frac{R_s}{L_d} & T_s p \omega_r \frac{L_q}{L_d} & 0 & 0 & 0 \\ -T_s p \omega_r \frac{L_d}{L_q} & 1 - T_s \frac{R_s}{L_q} & -T_s \frac{\Psi_f}{L_q} & 0 & 0 \\ 0 & a_{32} & 1 & 0 & -T_s \frac{1}{J} \\ 0 & 0 & p T_s & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (13)$$

$$\text{where: } a_{32} = T_s \frac{3}{2} \frac{p}{J} [\Psi_f - (L_q - L_d) i_d],$$

$$\mathbf{B}_k = \begin{bmatrix} T_s \frac{1}{L_d} \cos \gamma & T_s \frac{1}{L_d} \sin \gamma \\ -T_s \frac{1}{L_q} \sin \gamma & T_s \frac{1}{L_q} \cos \gamma \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad (14)$$

$$\mathbf{C}_k = \begin{bmatrix} \cos \gamma & -\sin \gamma & 0 & 0 & 0 \\ \sin \gamma & \cos \gamma & 0 & 0 & 0 \end{bmatrix}, \quad (15)$$

III. EKF STATE ESTIMATION

The Kalman filter is often applied during dissolving state estimation of dynamical system, disturbed by the known signals [1]. Kalman filter algorithm is used for estimating the parameters of linear system, but the PMSM model is nonlinear, so we can not use that filter in this case. Extended Kalman Filter is generalized algorithm, which can be used for nonlinear systems. The estimation is done upon undisturbed input signals (u_α , u_β for presented PMSM drive) and disturbed output signals (measured stator currents i_α , i_β) of a real nonlinear plant. As assumed, state space estimation is carried out in few steps at each computation cycle.

The EKF is an optimal estimator in the least-square sense for estimating the states of dynamic nonlinear system [1]. An linearization is build on assuming, that the state variables (7) are constant in one step of computation. We obtain new linearized matrices $\mathbf{A}_k(x_k)$, $\mathbf{B}_k(x_k)$ and $\mathbf{C}_k(x_k)$ valid only in one step. After this linearization the inner equation (9) gets a new form:

$$\frac{\partial \hat{\Phi}}{\partial \hat{x}} = \frac{\partial (\mathbf{A}_k(\hat{x}) \cdot \hat{x} + \mathbf{B}_k(\hat{x}) \cdot u)}{\partial \hat{x}}, \quad (17)$$

and output equation (10) the form:

$$\frac{\partial \hat{\mathbf{h}}}{\partial \hat{x}} = \frac{\partial (\mathbf{C}_k(\hat{x}) \cdot \hat{x})}{\partial \hat{x}}. \quad (18)$$

EKF algorithm consists of two main parts: measurement and time actualisation. Time actualisation equation:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + \mathbf{K}_k (y_k - \mathbf{C}_k \hat{x}_{k|k-1}) \quad (19)$$

predicts $\hat{x}_{k|k}$ state for instant time t_k . The correction proceeds on previous predicted state $\hat{x}_{k|k-1}$, using filter coefficient $\mathbf{K}_k$ (gain matrix) from t_{k-1} state and measured actual output vector y_k (currents). An error covariance is done by the following recursive relation:

$$\mathbf{P}_{k|k} = \mathbf{P}_{k|k-1} - \mathbf{K}_k \left. \frac{\partial \mathbf{h}}{\partial \hat{x}} \right|_{\hat{x}=\hat{x}_{k|k-1}} \mathbf{P}_{k|k-1}. \quad (20)$$

The object linearization (17) is attached to this group as well.

The second step – time actualisation, is described as the prediction of previous $\hat{x}_{k|k}$ state into a new state $\hat{x}_{k+1|k}$ used (8) in the form:

$$\hat{x}_{k+1|k} = \mathbf{A}_k \cdot \hat{x}_{k|k} + \mathbf{B}_k \cdot u_k. \quad (21)$$

The filter gain matrix is defined by:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \left. \frac{\partial \mathbf{h}^T}{\partial \hat{x}} \right|_{\hat{x}=\hat{x}_{k|k-1}} \left(\left. \frac{\partial \mathbf{h}}{\partial \hat{x}} \right|_{\hat{x}=\hat{x}_{k|k-1}} \mathbf{P}_{k|k-1} \left. \frac{\partial \mathbf{h}^T}{\partial \hat{x}} \right|_{\hat{x}=\hat{x}_{k|k-1}} + \mathbf{R} \right)^{-1} \quad (22)$$

The error covariance matrix is predicted by a relation:

$$\mathbf{P}_{k+1|k} = \left. \frac{\partial \hat{\Phi}}{\partial \hat{x}} \right|_{\hat{x}=\hat{x}_{k|k}} \mathbf{P}_{k|k} \left. \frac{\partial \hat{\Phi}^T}{\partial \hat{x}} \right|_{\hat{x}=\hat{x}_{k|k}} + \mathbf{Q}, \quad (23)$$

where $\mathbf{Q}$ and $\mathbf{R}$ are respectively: the covariance matrices of the system and measurement.

The whole algorithm is recursive by

$$\begin{cases} \hat{x}_{k|k-1} \rightarrow \hat{x}_{k+1|k} \\ \mathbf{P}_{k|k-1} \rightarrow \mathbf{P}_{k+1|k} \end{cases}. \quad (24)$$

An open question is the choice of the values of the matrices $\mathbf{Q}$ and $\mathbf{R}$. The change of covariance matrices effects both the dynamic and steady-state. The discussed matrices are simplified to diagonal form in order to eliminate state vector elements co-disturbance. Increasing particular values of $\mathbf{Q}$, it corresponds to stronger system noises and finally an instability in result of the strongest correction. Decreasing values of $\mathbf{Q}$, it corresponds to weaker correction and estimation state errors. Matrix $\mathbf{R}$ matches measurement noises. Intense existing measurement noises corresponds to necessity increasing. Basing on analytical rules described in [3] they were chosen and adjusted experimentally:

$$\mathbf{Q} = \begin{bmatrix} 2,3 \cdot 10^{-3} & 0 & 0 & 0 & 0 \\ 0 & 2,3 \cdot 10^{-3} & 0 & 0 & 0 \\ 0 & 0 & 1,138 & 0 & 0 \\ 0 & 0 & 0 & 10^{-4} & 0 \\ 0 & 0 & 0 & 0 & 3000 \end{bmatrix}, \quad (25)$$

$$\mathbf{R} = \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}. \quad (26)$$

That theoretical analysis of this consideration in [3] expanded to real plant by introducing scaling coefficients.

The presented algorithm does not require a precise choice of the initial values for the state vector $\hat{x}_{0|0}$ and the error covariance $\mathbf{P}_{0|0}$ matrix. It was proved during the experimental investigation, that $\mathbf{P}_{0|0}$ did not have a significant influence in the behaviour of EKF. The algorithm does not need the initial rotor position (non zero part of real state values) but it is possible to start the motor from the standstill place even if initial state vector of estimator is null.

IV. SENSORLESS CONTROL STRATEGY

Simplified structure of proposed sensorless control of Permanent Magnet Synchronous Motor is shown in figure 1. Control diagram consists of PI current controllers subordinated to speed controller and adequate frame coordinate transformations. The main state of presented system is an estimator, realized by Kalman Filter theory operation.

The PI speed controller feeds current i_d^* in q axis in order to keep Field Oriented Control [4]. The demanded current is computing by using the difference between requested speed (ω_r^*) and speed ($\hat{\omega}_r$) estimated by Kalman filter. Motor operating does not require the field weakening, as assumed. Therefore desired current i_d^* in d axis is maintained to zero. These signals are inputs of PI current controllers, which provides desired voltages in dq reference frame. Basing on the shaft position γ , voltages are converted into the stationary two axis frame ($\alpha\beta$) and send to control Pulse Width Modulation inverter.

V. EXPERIMENTAL RESULTS

A. Laboratory Setup

For experimental verification of the proposed estimation method, a laboratory setup has been constructed. It consists of the surface mounted magnets synchronous motor, supplied from the three phase power IGBT inverter. This motor is

coupled via stiff shaft with the second twin motor supplied from industrial controller (DIGITAX made by Control Techniques).

The voltage and currents signals are adjusted and sampled simultaneously with 12-bit A/D converters. The rotor position is measured by 3600 pulse per revolution incremental encoder, only for comparing.

The presented algorithm was implemented on Analog Devices *Sharc 21061* Digital Signal Processor [6]. The code has been mainly written in C language, except low-level procedures written directly in assembler. However the PWM generator was realised with additional motion coprocessor *ADMC201* used for fast hardware pulses generating. A real position is computed using *Altera Flex* and sent to DSP by parallel memory fields. An execution time of EKF algorithm is about $130\mu s$ while a control algorithm takes $30\mu s$. The sampling time is $T_s = 200\mu s$ and is equal to current and speed control. The System contains also one four-channel D/A converter, that was built in for auxiliary outputs, helpful in state visualisation.

B. Obtained Results

At the first part of investigation, it was focused on controlled system behaviour by reference speed (ω_r^*) excitation without additional load torque. This type of reference signal has such significant stages for estimator behaviour like: zero signal with no initial values in estimation vector, step signal and reversal. The maximum module of reference speed is 1000 rev/min ($1/3$ of maximum speed). Results of working are presented on fig. 2. The position error is counted by:

$$\Delta\gamma = \hat{\gamma} - \gamma \quad (27)$$

This figure shows some kind of error that occurs when speed is suddenly changed. The source of errors stays in the differences between motor model and real plant, particularly at current limitation in current controller.

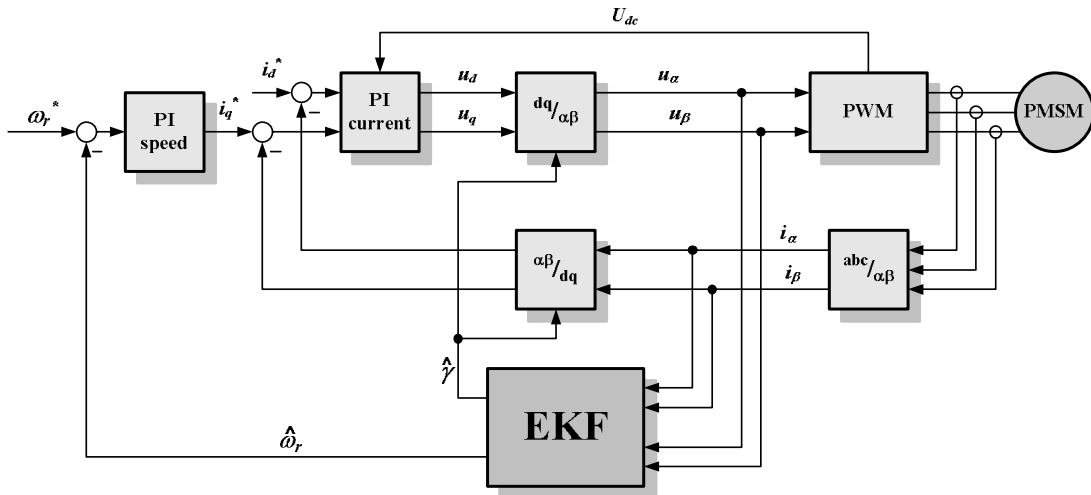


Fig. 1. Control strategy of sensorless PMSM drive

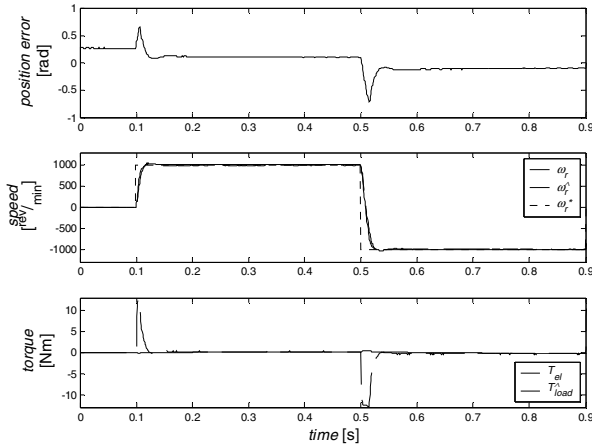


Fig. 2. Experimental trial of system during motor start to 1000 $[\text{rev}/\text{min}]$ and reversal to -1000 $[\text{rev}/\text{min}]$.

Figure 3 shows better results with exclusion of unwanted current limitation. There are established value of maximum reference speed derivative equal $6 \cdot 10^4 \text{ rev}/\text{min}^2$, there is about a half of the maximum torque with nominal moment of inertia.

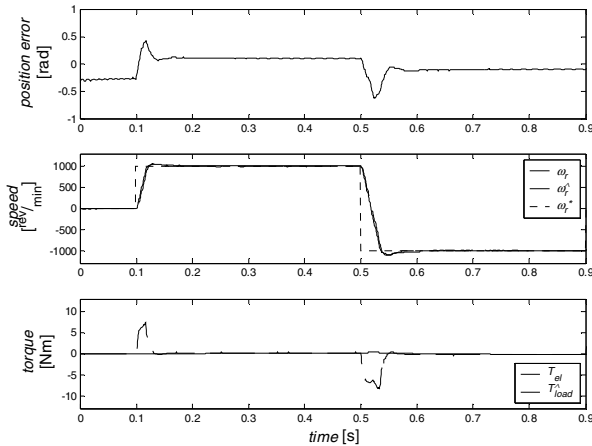


Fig. 3. Experimental trial of system during motor start to 1000 $[\text{rev}/\text{min}]$ and reversal to -1000 $[\text{rev}/\text{min}]$ with speed derivative limited.

Figure 4 presents the response of the PMSM to a load step (50% rated torque) at a motor speed of 1000 rev/min . In steady state, the estimated load equals to electromechanical torque. Erroneous load torque dynamical reconstruction declined fast. This influence is small compared to a potential load variation.

In the author opinion the construction of PMSM issues the apparent systematic position error.

VI. CONCLUSION

This paper has proposed a hardware and software strategy of sensorless PMSM control. A systematical and analytical approach for developing the algorithm was presented. The

discrete Extended Kalman Filter was found to be well suited to the speed, position and load torque estimation of PMSM drive. The proposed application has been validated using real time system. The results have shown that the proposed

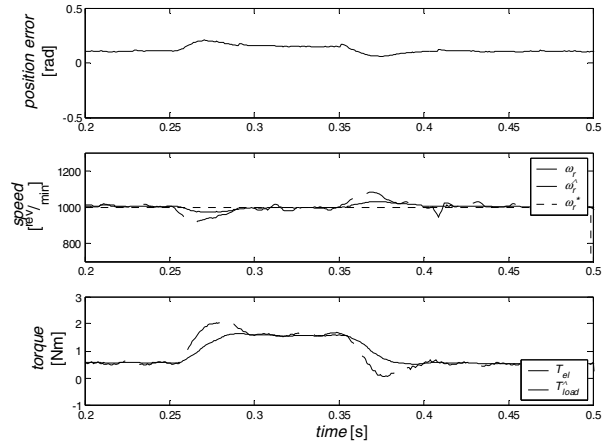


Fig. 4. Experimental trial of system during load torque acting .

control strategy had a good dynamic response in a wide speed range and load torque. The development of an experimental drive system is under way.

VII. ACKNOWLEDGE

This work is supported by the KBN (the Polish Committee for Scientific Research) under Grant 3 T10A 026 28.

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